Roll No. .....

MAI–101

M. B. A. (AI & DS) (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

MANAGEMENT CONCEPTS AND
ORGANIZATIONAL BEHAVIOR

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections—Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) and (b) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section–A

1. (a) Evaluate implications of Elton Mayo's contributions, for managing modern organizations. (CO3/CO4/CO5)

OR

(b) Is your college Director a manager ? Discuss in terms of Fayol's Managerial functions, Mintzberg's roles and Katz's skills. (CO3/CO4/CO5)

P. T. O.

2. (a) Analyze relevance of classical approach in the contemporary global business scenario. (CO3/CO4/CO5)

OR

(b) Analyze importance of ESG policy for organizations. (CO3/CO4)

3. (a) Explain different phases of design thinking. (CO1/CO2/CO3)

OR

(b) Explain the potential Global competitive challenges faced by digital managers due to IR 5.0. (CO3/CO4/CO5)

Section–B

4. **Industry Insight Analysis :**

Enterprises' initial efforts to address sustainability issues have been largely driven by public relations or brand defense—with less regard for meaningful outcomes. More recent efforts, such as reducing emissions in operations or manufacturing, are producing real but incremental improvement at the fringes of the business. Now it's time to take the next step to achieve tangible outcomes by focusing on the human element by operationalzing people-enablers in the organization, planning for the right technical and soft or human skills, making work better for humans; and designing for human sustainability. This is where fundamental and lasting change takes root and outcomes can be achieved. In the Deloitte 2023 Global Human Capital Trends survey, 84% of respondents acknowledge that understanding the impact of sustainability on their organization and defining ownership for driving progress and outcomes is important to their organizations' success. Yet only 21% believe that their organizations are very ready to address such issues. Many organizations have begun to value not only environmental

stewardship, but also human sustainability, or the creation of value for current and future workers and, more broadly, human beings and society. These organizations unleash individuals' potential and help them become healthier, more skilled, and more connected to a sense of purpose and belonging. In the Deloitte Global 2022 Gen Z and Millennial survey, nearly half of respondents said they have personally put pressure on organizations to act on sustainability topics. Thirty percent of workers in a 2021 Deloitte external survey said they would consider switching jobs to work for a company that is more environmentally sustainable. (Courtesy : Deloitte) Based on the excerpts & your industry insights reading, answer these questions :

(a) Human sustainability rises to the fore of organizations' sustainability strategies. Comment. (CO3/CO4)

OR

(b) Develop a blueprint to develop policies for promoting sustainability. (CO3/CO4)

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MAI–102

M. B. A. (AI & DS) (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

MANAGERIAL ECONOMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections—Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) and (b) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section–A

1. (a) Illustrate the circular flow model of an economy and elaborate on the interaction among various entities involved therein. (CO1/CO2)

OR

(b) Analyze a real-world scenario and identify the central economic problems faced by a country or a company. How does economics help address these problems ? (CO1/CO2)

P. T. O.

2. (a) Discuss the properties of indifference curves with reasons and diagrams for illustration. (CO2/CO3)

OR

(b) Why does the demand curve have a negative slope ? Analyze and support your answer with reasons. (CO2/CO3)

3. (a) Explain the concept of price elasticity of demand and supply. How does price elasticity impact pricing and revenue decisions for businesses ? (CO2/CO3)

OR

(b) Analyse how changes in supply and demand lead to shifts in market equilibrium. Provide an example of a product or service of your choice, and explain how external factors can create shortages or surpluses. Illustrate with diagram. (CO3/CO4)

Section–B

4. **Case Study :**

"The Coffee Shop Dilemma : Demand, Supply, and Utility Analysis"

Scenario :

You are the manager of a popular coffee shop located in a bustling downtown area. Your coffee shop offers a variety of coffee blends, pastries, and snacks. Recently, you noticed a change in customer preferences and external factors that are affecting your business. You decide to apply economic principles to analyse the situation and make informed decisions.

Background :

Over the past few months, the price of coffee beans has increased due to adverse weather conditions in major coffee-producing regions. Simultaneously, a new health trend promoting herbal teas and healthy snacks has emerged in your area. This has led to a decline in the demand for traditional coffee beverages and an increase in the demand for herbal teas and low-calorie snacks.

(a) Apply the concepts of demand and supply to analyse the situation faced by the coffee shop. How have changes in the price of coffee beans and shifts in customer preferences impacted the demand and supply for your coffee products ? Provide specific examples and discuss the implications for pricing and inventory management. (CO5/CO6)

OR

(b) As the coffee shop manager, you need to make strategic decisions to adapt to the changing market conditions. Based on your analysis of demand, supply, and customer preferences, outline a set of recommendations for the coffee shop. How can you adjust your product offerings, pricing strategy, and marketing efforts to ensure the continued success of your business ? Consider the concept of utility in your recommendations. (CO5/CO6)

Roll No. .....

MAI–103

M. B. A. (AI & DS) (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

MARKETING MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) Define marketing how the concept of marketing has over a period of time. (CO1)

OR

(b) Discuss need and importance of marketing in the 21st century. (CO2)

2. (a) How would you analise the factors while a selecting the market for doing the business for a firm ? (CO1)

P. T. O.

OR

(b) We generally talk about business environment, is it different from marketing environment ? Comment. (CO2)

3. (a) Comment on the consumer decision-making process and its significance. (CO3)

OR

(b) Throw some light on Segmentation, Targeting and Positioning. (CO3)

Section—B

4. **Case Study :** (20 Marks)

Film 'Sanju'

Sanjay Dutt is a popular name in Indian cinema. He is the son of legendary actor Sunil Dutt and Nargis Sanjay enjoys a high screen presence in today's Hindi cinema. But in the past, Sanjay was into drugs and was very violent. Sanjay was accused of being involved in Bombay bomb blast for which he was jailed for sometime. He underwent rehabilitation and now spends quality time with his friends and family. Sanjay is associated with few non-governmental organizations and participates in road shows, seminars, conferences and advertises against abuse of drugs. He is well known for his comic roles also, like the role-played in a movie called "Munna Bhai MBBS". There is a biopic named after him as "Sanju" in which Ranbir Kapoor essayed the life of Sanjay Dutt.

It is necessary to add humour to communication against drug abuse and Sanjay feels that he fits into such a humour character who can motivate people against drugs. He took this responsibility and spoke to people how he came out of drugs with lot of humour and fun. He visits clubs, schools and colleges and talks about the don'ts about drugs.

Questions :

(i) What environmental factors influence in making a successful comedy cinema–whether it's the actor's ability or the content of communication ? (CO3)

(ii) How do you think celebrity endorsement exercise like this should target normal people, children who are more vulnerable to drugs and to addicts when each one of them have different entertainment needs ? (CO2)

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MAI–104

M. B. A. (AI & DS) (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

PROGRAMMING IN PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) What are data types in Python, and how do you determine the data type of a variable ? (CO1)

OR

(b) Explain the concept of a Python list and provide an example of how to create and manipulate one. (CO3)

P. T. O.

2. (a) What is a Python dictionary, and how is it different from a list ? (CO3)

OR

(b) What is a Python module, and how do you import and use functions or variables from a module ? (CO1)

3. (a) Explain the concept of object-oriented programming (OOP) in Python and provide an example of creating a class and an object. (CO2)

OR

(b) What will the following Python code print ? (CO1/CO3)

```
numbers = [1, 2, 3, 4, 5]
for num in numbers :
print (num * 2)
```

Section—B

4. **Case Study :** (20 Marks)

Problem Statement : Develop a Python program that converts temperatures between Celsius and Fahrenheit. Describe how you would take user input, perform the conversion, and display the result. (CO1/CO3)

OR

Problem Statement : Imagine you are tasked with creating a basic calculator program in Python. Describe how you would design and implement this program, allowing users to perform addition, subtraction, multiplication, and division operations. (CO1/CO3)

Roll No. .....

MAI–105

M. B. A. (AI & DS) (FIRST SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

DATA WAREHOUSE AND DATAMINING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) Explain the fundamental concepts of data warehousing. Discuss the key components of a data warehouse architecture and their roles in facilitating effective data management and analysis. Provide examples to illustrate your points. (CO3)

OR

(b) Describe the importance of data modeling in the context of data warehousing. Compare and contrast different data modeling techniques such as star schema and snowflake schema. Discuss the factors that influence the choice of a particular data modeling approach. (CO1)

P. T. O.

2. (a) Define ETL (Extract, Transform, Load) processes in the context of data warehousing. Discuss the significance of ETL in data integration and data quality. Provide a step-by-step explanation of the ETL process and the challenges involved in each phase. (CO1/CO2)

OR

(b) Explain the steps involved in designing and implementing a data warehouse. Discuss the various design considerations, including schema design, indexing, and partitioning. Highlight best practices for optimizing query performance in a data warehouse. (CO1)

3. (a) Provide an overview of popular data warehousing tools and technologies. Compare and contrast traditional on-premises data warehousing solutions with cloud-based data warehousing services. Discuss the advantages and disadvantages of each approach. (CO3)

OR

(b) Identify and discuss the common challenges and emerging trends in the field of data warehousing. Explain how advancements in technologies like artificial intelligence and big data are impacting the evolution of data warehousing practices. Provide examples of real-world use cases to illustrate these challenges and trends. (CO2/CO3)

Section—B

4. **Case Study :** (20 Marks)

XYZ Retail Inc. is a thriving retail chain with numerous stores across the country. Over the years, the company has gathered a substantial amount of

sales data from its various stores, but it's becoming increasingly challenging to analyze this data effectively. The existing system relies on multiple databases, spreadsheets, and reports generated from various sources, making it difficult to obtain a unified view of sales performance.

Challenges :

Data Fragmentation : Sales data is scattered across different locations and formats, making it hard to consolidate and analyze efficiently.

Performance Issues : Generating complex sales reports takes a long time, and the reports often lack real-time accuracy.

Data Quality : There are concerns about data accuracy, consistency, and the presence of redundant information.

Scalability : As the company expands, the volume of data continues to grow, which further complicates data management and analysis.

(i) How will you address the above challenges ? (CO1/CO2)

(ii) Give the advantageous of solution provides you. (CO1/CO2)

Roll No. .....

MAI–106

M. B. A. (AI & DS) (FIRST SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FUNDAMENTAL OF DATA SCIENCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) What is the fundamental difference between descriptive and inferential statistics in the context of data analysis ? (CO1/CO2)

OR

(b) Explain the concept of variance and standard deviation in statistics. How are they related to each other, and what do they indicate about the spread or dispersion of data ? (CO1/CO2)

P. T. O.

2. Define the following :

(a) What is the significance of the normal distribution in statistics ? How does it relate to the concept of the bell curve, and why is it commonly used in various statistical analyses ? (CO1/CO2)

OR

(b) Describe the concept of sampling in statistics. What is a random sample, and why is it important for making valid statistical inferences ? (CO1/CO2)

3. (a) What is a hypothesis test in statistics, and why is it used ? Walk through the basic steps involved in conducting a hypothesis test.

OR

(b) Write an SQL code to get a list of departments in a company (Department_ID) with the average salaries of employees in the department. The code should also incorporate a condition that only those departments should be part of output where the average salary of the department is >5,000.

Note : The database from where you have to fetch the data is titled 'Employees' and has fields like 'Department_ID and 'Salary'.

(CO2/3/4/5)

Section—B

4. **Case Study :** (20 Marks)

Let us assume you are working as a data scientist with a firm. A client of the firm wants you to develop an AI-based product/application to monitor

the farmland of the company. Considering the scenario discuss the following : (CO3, 4, 5)

(a) The key skills of professionals you may need to deliver the project to the client.

(b) A structured approach you will take to execute the project.

(c) Illustrate with examples the data collection templates you may design to collect the primary data in the database.